ChanningCGDS

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Welcome

Package: ChanningCGDS **Authors:** Vincent Carey [aut, cre] **Compiled:** 2025-07-16 **Package version:** 0.98.2 **R version:** R version 4.5.1 (2025-06-13) **BioC version:** 3.22 **License:** MIT + file LICENSE

This is the landing page of the `BiocBook` entitled ….

This book introduces the reader to ….

Docker image

A `Docker` image built from this repository is available here:

ghcr.io/vjcitn/channingcgds

Get started now

You can get access to all the packages used in this book in $<$ 1 minute, using this command in a terminal:

Listing 0.1 `bash`

```
docker run -it ghcr.io/vjcitn/channingcgds:devel R
```

RStudio Server

An RStudio Server instance can be initiated from the `Docker` image as follows:

Listing 0.2 `bash`

```bash
docker run \
    --volume <local_folder>:<destination_folder> \
    -e PASSWORD=OHCA \
    -p 8787:8787 \
    ghcr.io/vjcitn/channingcgds:devel
```

The initiated RStudio Server instance will be available at https://localhost:8787.

Session info

Click to expand

1 Road map

2 Computing platform concepts

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3 Storage concepts for cohorts and experiments

4 Data structure concepts for genomic analysis

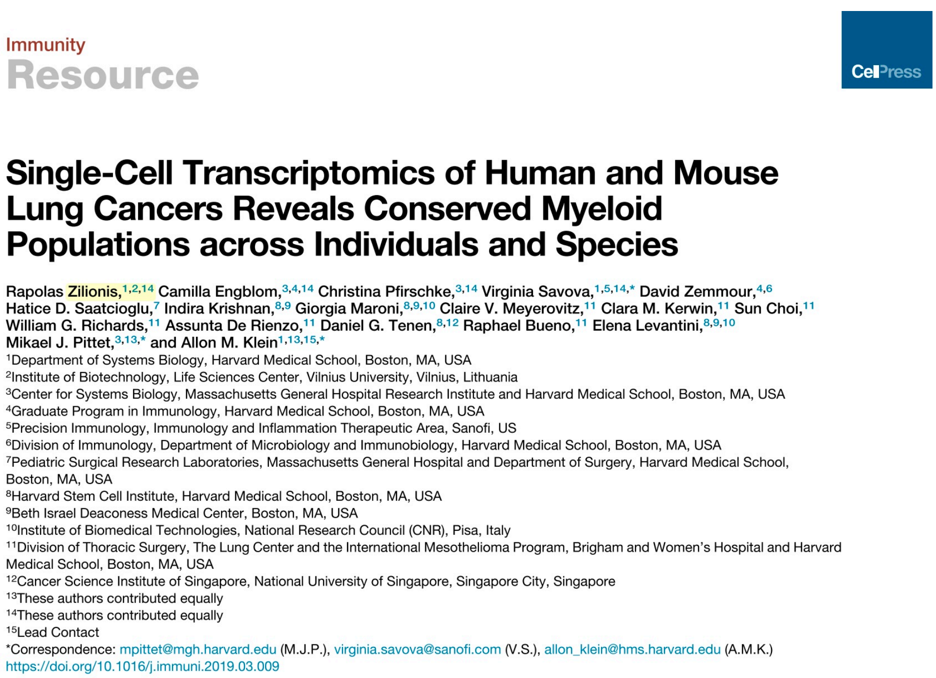
4.1 Self-descriptive data structures

4.1.1 Example: bulk RNA-seq analysis of smooth muscle treated with dexamethasone

rnaSeqGene workflow

pubmed link to original study

4.1.2 Example: Zilionis Single-Cell RNA-seq in lung



Immunity

Resource

CellPress

Single-Cell Transcriptomics of Human and Mouse Lung Cancers Reveals Conserved Myeloid Populations across Individuals and Species

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https://doi.org/10.1016/j.immuni.2019.03.009

Figure 4.1: Title of paper

scRNAseq

```
library(scRNAseq)
library(SingleCellExperiment)
library(ggplot2)
```

```
zil = ZilionisLungData()
```

```
zil
## class: SingleCellExperiment
## dim: 41861 173954
## metadata(0):
## assays(1): counts
## rownames(41861): 5S_rRNA 5_8S_rRNA ... snosnR66 uc_338
## rowData names(0):
## colnames(173954): bcIIOD bcHTNA ... bcELDH bcFGGM
## colData names(16): Library Barcode ... used_in_NSCLC_non_immune
##   used_in_blood
## reducedDimNames(5): SPRING_NSCLC_all_cells
##   SPRING_NSCLC_and_blood_immune SPRING_NSCLC_immune
##   SPRING_NSCLC_non_immune SPRING_blood
## mainExpName: NULL
## altExpNames(0):
```

4.1.2.1 Why is this "self-descriptive"?

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4.1.2.2 Excursus: classes and methods

SingleCellExperiment

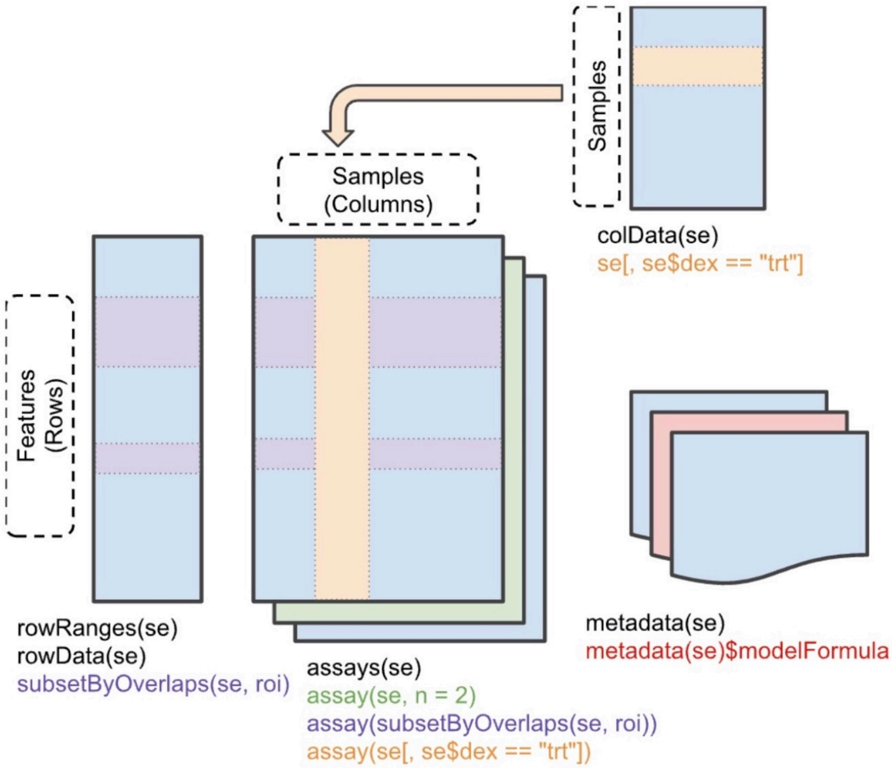

Figure 4.2: SummarizedExperiment schematic

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```
class(zil)
##  [1] "SingleCellExperiment"
##  attr(,"package")
```

```
## [1] "SingleCellExperiment"
is(zil, "SummarizedExperiment")
## [1] TRUE
```

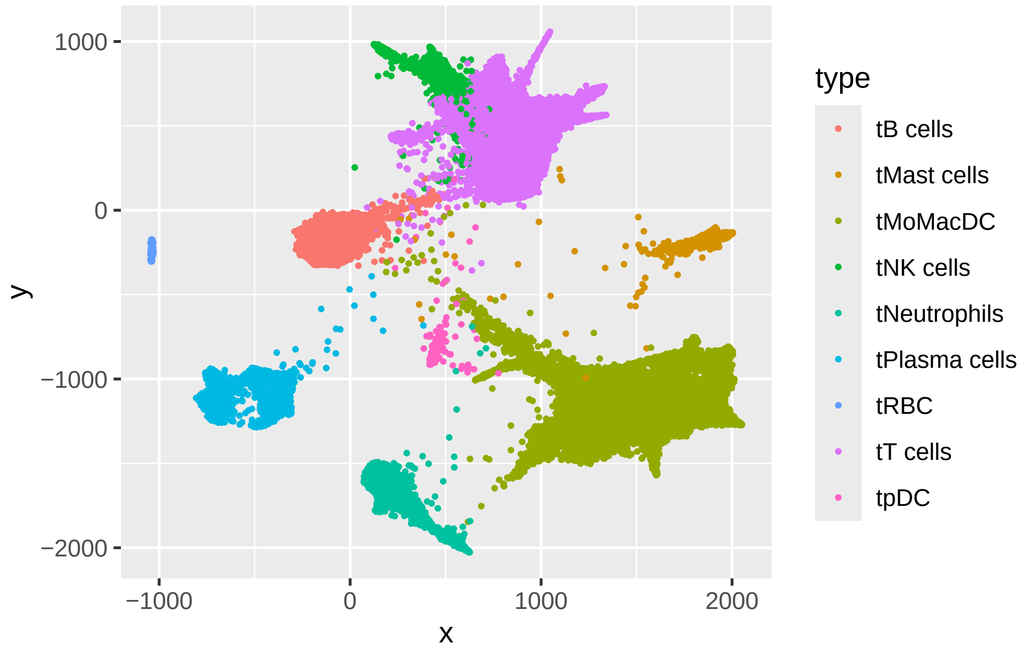
"S4 object-oriented programming"

```
allm = methods(class="SummarizedExperiment")
grep("assay,", allm, value=TRUE)
## [1] "assay,SummarizedExperiment,character-method"
## [2] "assay,SummarizedExperiment,missing-method"
## [3] "assay,SummarizedExperiment,numeric-method"
```

4.1.2.2.1 How does this represent useful binding of metadata to data?

```
table(zil$Patient, zil$Tissue)
##
##      blood tumor
##   p1 12763 21243
##   p2 15826 14873
##   p3  8165 30794
##   p4  4906 11907
##   p5     0 14049
##   p6  3964 11300
##   p7  6981 17183
```

```
library(ggplot2)
rd = as.data.frame(reducedDims(zil)[[3]])
rd = cbind(rd, type = zil$`Major cell type`)
rdp = na.omit(rd) # many cells not used
ggplot(rdp, aes(x=x, y=y, colour=type)) + geom_point(size=.5)
```



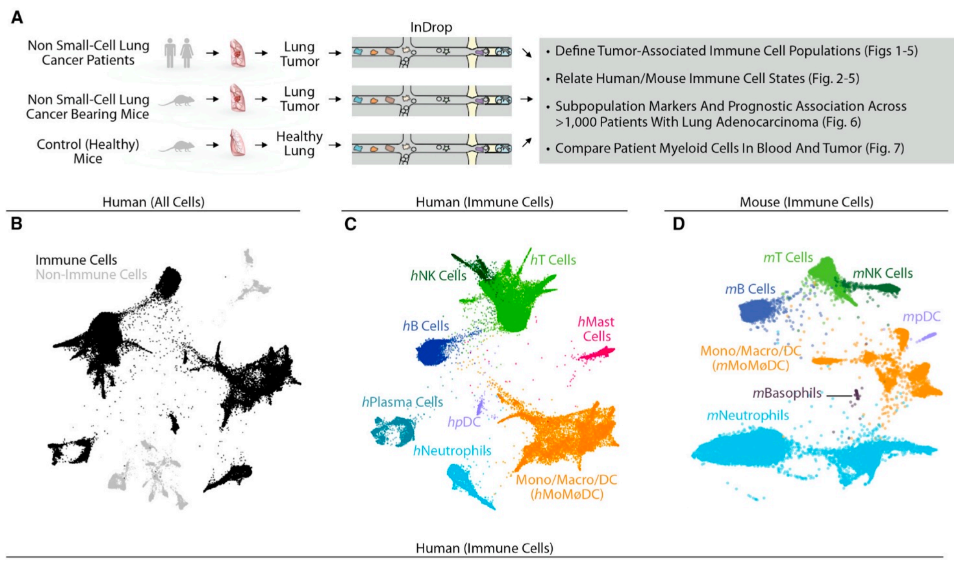


Figure 4.3: Fig 1 excerpt

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5 Analytic software and workflows